

Bank Marketing

The Insight Lab Team 2b

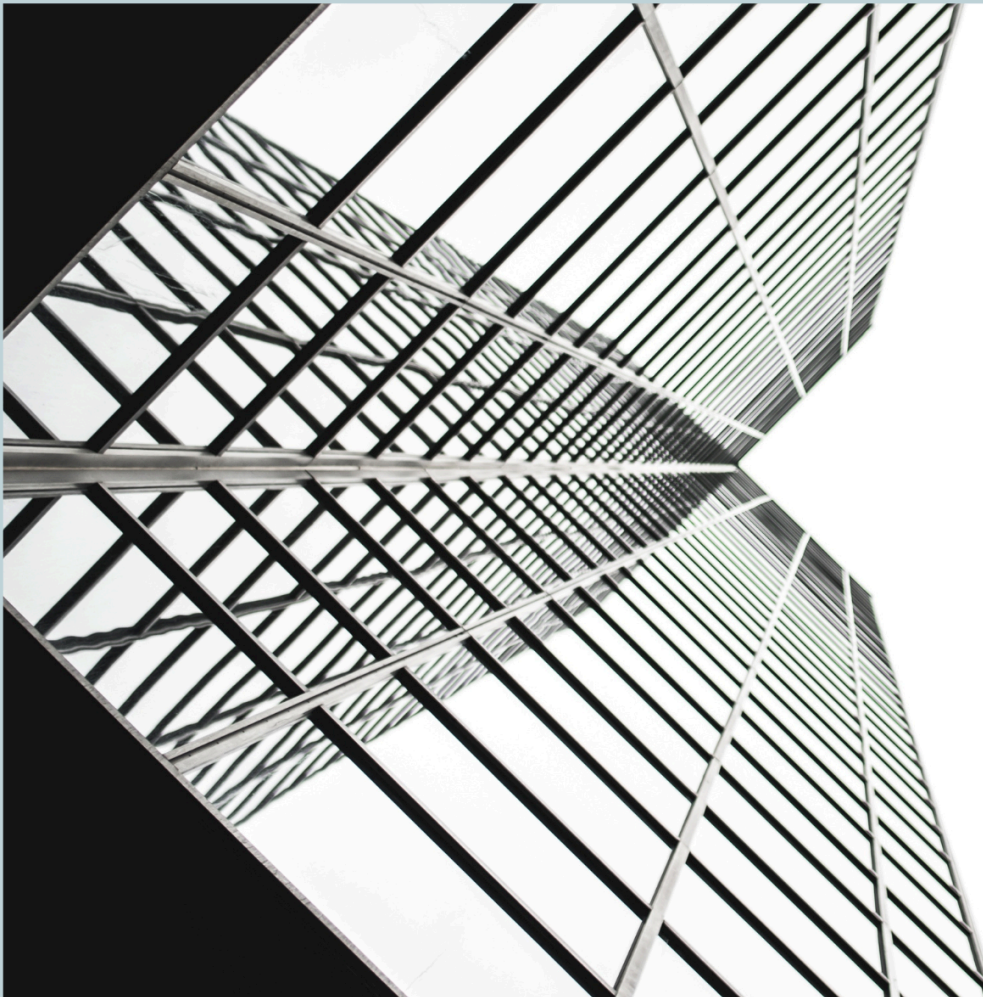
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1. Executive Summary

Banks are known to collect nearly all details of their customers. Employment, financial stability, loan history, and previous actions, which are customer data collected by most banks, can be used to learn more about the habits of customers, as well as what will attract potential customers. Approaching marketing campaigns with the adequate machine learning models will help prioritize clients with the higher likelihood of subscribing to banking services.

Based on data collected from a Portuguese banking institution from 2008-2010, our data focused on customer attributes that would lead to enrollment in the bank's services after receiving promotional materials via phone call. The goal of the project is to find the machine learning model that is the most effective in predicting customer subscription rate.

Based on both decision tree Classification and logistic regression algorithms, as well as sampling and hypertuning parameters via SMOTE, the model that predicted with the highest accuracy was the SMOTE hyperparameter tuned logistic regression model. This model, which was chosen due to its higher accuracy, has a stratified "yes" accuracy of 0.799 and a ROC AUC score of 0.802, which provides reasonable accuracy in predicting if a customer that was called was willing to enroll in the banking service.

2. Introduction

Studies have shown that investing is at an all time high. As many investment opportunities arise, the most common type of investment is low-risk investments. The majority of these opportunities are easily accessible by a personal bank however, not everyone that is reached out by the bank's marketing campaign will be willing to invest. Factors such as job category, whether they are retired or unemployed, loan amount, any loans or not, and previous historical campaign contacts, their previous answers being 'Yes' or 'No', will be strong predictors of whether customers will say 'Yes' or 'No' to such opportunities this time around.

Marketing finance products such as low risk investments is such an important priority for banks. As much effort is created, not every customer will be interested in subscribing to term deposits. Contacting customers can be very time consuming as well as a financial waste when most answers turn out to be 'No'. Understanding why a customer will answer with 'Yes' will create a big improvement in time management and overall efficiency in the bank's marketing strategy.

Our project focuses on predicting which customers are most likely to subscribe into the bank's term deposit and what are key factors that influence their decision. We will focus on demographics, financial characteristics, and outcomes from previous marketing campaigns. In order to gather the proper recommendations, models such as logistic regression and decision trees were used. After processing the data, we compared both results to ensure we got the best understanding and recommendations for improving the bank's marketing strategy.

Our project uses one of the four datasets available to use from a local bank in Portugal, where marketing campaigns were conducted via phone calls, with instances in which contact was made more than once to the same client in order to obtain their final response. We decided to keep the main overall dataset with the entirety of its information to develop a correct analysis determining the factors, recommendations and steps to take for this bank. The data set contains 41,188 examples that have been organized by date from May 2008 to November 2010 and was accessed via UCI's Machine Learning Repository.

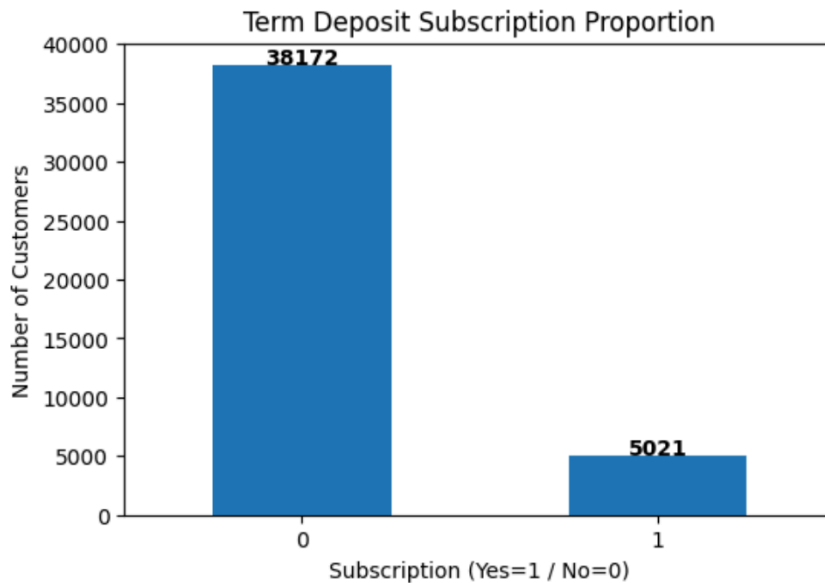
3. Data

In our bank marketing dataset from Portugal, there are 41,188 observations each representing a customer who was contacted in part of the phone term deposit investment. The data as mentioned earlier was collected by this bank between the dates of May 2008 and November 2010. In each record per customer, contains customer demographics, financial information, and details about previous campaign participations.

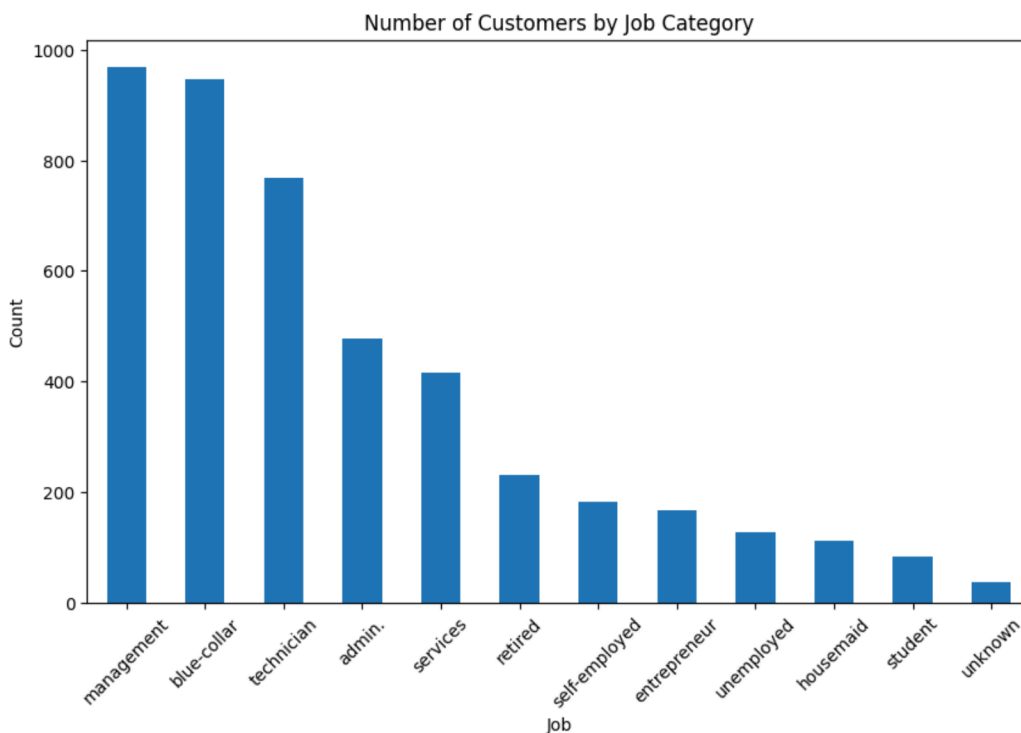
The variables included for this analysis are:

- Age: Customer's current age
- Job: Employment type
- Marital: Marital status
- Education: Education level
- Default: Whether the customer is default of credit ('yes'/'no')
- Balance: Customers average yearly balance in bank (euros)
- Housing: Whether customer has housing ('yes'/'no')
- Loan: Whether customer has any sorts of personal loans ('yes'/'no')
- Duration: Duration of last call
- Campaign: Number of contacts in previous campaign
- Previous: Number of previous contacts
- Pdays: Days since last contact in previous campaign
- Previous: Number of previous contacts

The variables above are a mix of categorical variables as well as continuous variables such as age, balance, pdays, and duration. Above all, our target variable, 'Term Deposit Subscription', is a binary variable with 'Yes' and 'No' as possible outcomes.



The “Term Deposit Subscription Proportion” chart displays the portion of customers who subscribed to the banking institution after the calls, with ‘no’ mapped to 0 and ‘yes’ mapped to 1. Here we can see the clear imbalance between the target variable with most customers saying ‘no’. This imbalance leads us to notice that we will need to make adjustments in order to improve accuracy.



The “Number of Customers by Job Category” graph shows us just how many categories in the workforce there are. This demonstrates that we will be needing to incorporate a better distribution that will be less complex in order for there to not be an incorrect outcome due to the large imbalance.

4. Analysis

Logistic Regression and Decision Tree Classifier

In order to increase the probability of predicting whether a customer would subscribe to a term deposit, we developed two machine learning models to our data, logistic regression and decision tree classifier. For logistic regression, we decided to use this method as it is simple, straightforward, interpretable, and a great fit when it comes to binary classification problems such as ‘yes’/‘no’. As for decision tree classifiers, this model provides the ability to capture non linear relationships and interactions between different variables. It overall can become a simple method to demonstrate visualizations and create accurate interpretations. Overall in both methods, they allow for tuning via pruning, and variable weightage.

Metrics for Evaluation

In our data, we predicted how well each model did with our modifications via four scores: overall accuracy, stratified accuracy, f1-score, and ROC AUC scores. A brief explanation of each metric is explained below:

- Overall Accuracy: Determines the overall predictions that the model made correctly over all predictions made. Is more useful when the data has an equal proportion of 0s and 1s, as well is doing around the same rate with both true positives and true negative
- Stratified Accuracy: Splits the accuracy measure by 0 and 1, and is more useful for checking if the model is predicting well for true positives and true negatives
- F-1 Score: Used for more unbalanced data, it uses both precision and recall in order to evaluate the model, which takes into account the accuracy of predicted positives as well as coverage of the dataset’s true positives in the prediction. Is good at predictions if you set a threshold of the model at a certain value.
- ROC AUC Score: The integral of the ROC curve, which compares the true positive rate to the false positive rate, based on the threshold of the model. Is useful for skewed proportions of data, where multiple thresholds for the data should be tested

Since the model is trained on data that is disproportionate in the number of 0’s and 1’s in the data, overall accuracy cannot be used to determine the model’s viability, since we care more about customers willing to subscribe to the bank (1). Additionally, the ROC AUC score is more robust than the f1 score since it evaluates the data performance data at all potential thresholds instead of the one set by the user when coding the model (in our data 0.5 and above rounded up to 1, while below 0.5 rounded to 0). Therefore the most prominent scores we evaluate our model on is ROC AUC and stratified accuracy, with a lesser focus on f-1 score.

For all models that were created other than the hypertuned SMOTE models, predictions were made with a single fold classification, as well as a 10-fold cross-validation with all four metrics listed above recorded. For the SMOTE hypertuned model, both the decision tree classifier and logistic regression models worked the best with 5-fold cross-validation, so we only ran the optimized tests. Additionally, each model had a confusion matrix in order to measure raw counts, as well as feature importances evaluated (which were scaled in the logistic regression model in order to be comparable to the decision tree model), in order to understand what features contributed to predicting the model the most accurately.

5. Results

All our confusion matrices are found in the technical appendix, as well as feature importance charts, our code, and the link to the data set that we used.

Benchmark 1: Numerical Data

Benchmark 1 was the most similar to the original data, with the only modifications being all categorical variables that were dropped (other than the boolean y, which was converted to numerical data). Therefore, it also has the least amount of features to be evaluated on.

Metrics (10 fold cross validation)	decision tree	logistic regression
Accuracy	0.853 +/- 0.004	0.889 +/- 0.003
F1-Score	0.395 +/- 0.014	0.265 +/- 0.023
ROC AUC	0.661 +/- 0.009	0.578 +/- 0.009
Class 0 Accuracy	0.911 +/- 0.005	0.984 +/- 0.002
Class 1 Accuracy	0.411 +/- 0.019	0.171 +/- 0.018
Top 3 Important Features	duration, balance, day	duration, campaign, pdays

Between the decision tree and the logistic regression models that were calculated, the decision tree model does significantly better than the logistic regression in terms of ROC AUC, class 1 accuracy, and f1 scores. However, the large discrepancy between class 0 and class 1 accuracy shows that the models are being considered accurate since there are more instances of 0 in the actual data. This means that the model is either focusing on class 0 predictors (rather than class 1 predictors that we are trying to optimize for), or broadly classifying most instances under class 0 instead of evaluating data features, and needs to be processed further.

In terms of feature importance, both models evaluated that the duration of the call was the most predictive factor in if the person was going to subscribe to the banking institution, but the logistic model was more likely to weigh in more campaign related attributes compared to the decision tree classifier, who evaluated more based on the attributes of the caller.

Benchmark 2: Cleaned Data

Benchmark 2 was cleaning categorical data in order to make it usable for the data models, therefore increasing the variables that were used to calculate the data. Additionally, rows that did not have information were removed.

See the table below for the exact variables modified:

Preprocessing Action	Variables
Columns Deleted	- contact - day - month - poutcome
Rows dropped if “unknown”	- job - marital - education
Binning	- job: students were classified as unemployed - job: all jobs that were there classified under employed
Change Yes and No to 1 and 0	- y - housing - loan - default
Adding Dummy Variables for Categorical Variables	- job - marital - education

Preprocessing of variables conducted for the models for Benchmark 2. Both SMOTE models were sampled based on the data that was cleaned in Benchmark 2.

Metrics (10 fold cross validation)	decision tree	logistic regression
Accuracy	0.852 +/- 0.003	0.892 +/- 0.002

F1 Score	0.388 +/- 0.014	0.316 +/- 0.016
ROC AUC	0.657 +/- 0.010	0.598 +/- 0.006
Class 0 Accuracy	0.911 +/- 0.005	0.981 +/- 0.002
Class 1 Accuracy	0.403 +/- 0.023	0.215 +/- 0.013
Top 3 Important Features	duration, balance, age	duration, housing, campaign

The decision tree model performs better than its logistic counterpart in benchmark 2, with a better f1 score, ROC AUC, and class 1 accuracy. However, the logistic regression model has increased its predictive abilities for predicting subscription than from benchmark 1, with a higher f1, ROC AUC and class 1 score than benchmark 1. However, for the decision tree classifier model benchmark 1 was slightly better at predicting f1 score, ROC AUC, and class 1 accuracy, making the decision tree model in benchmark 1 the best so far.

Feature sampling was also similar between benchmark 2 and 1, with duration of the calls seeming to be the most important feature for both the decision tree model and the logistic regression model.

SMOTE Sampling

In the first batch of SMOTE sampling, the data was resampled so there was an even amount of 0s and 1s in the true data which was tested, making the data more likely to look at the factors that would make a person subscribe to the banking institution instead of passing. This data uses the cleaned data from benchmark 2, and resamples the data via making new samples based on data points.

Metrics (10 fold cross validation)	decision tree	logistic regression
Accuracy	0.828 +/- 0.004	0.807 +/- 0.007
F1 Score	0.393 +/- 0.016	0.485 +/- 0.012
ROC AUC	0.676 +/- 0.013	0.796 +/- 0.009
Class 0 Accuracy	0.874 +/- 0.004	0.811 +/- 0.008
Class 1 Accuracy	0.479 +/- 0.026	0.781 +/- 0.018
Top 3 Important Features	duration, balance, housing	duration, housing, marital_married

Both the decision tree classifier and the logistic regression model do better in our important measurements than the unprocessed data that we used for benchmark 1 and 2. However, what stands out is that the logistic regression model becomes drastically better, with a class 1 accuracy that goes up by more than .55 from 0.215 to 0.789, which makes it the most effective model of the 3 discussed so far, and emphasizes the need for balanced data when training machine learning models.

Feature testing for both the decision tree and the logistic regression continue to rely on duration, which follows suit since the new data added is similar to the data collected in the original data set. For both of the models, other than the duration, most of the predictive features tend towards customer attributes than the marketing techniques used.

SMOTE Sampling and Hyperparameter Tuning

In the last SMOTE testing, the models were optimized using GridSearch CV in order to find the best parameters for optimizing for occurrences of subscription. Each model had their own optimized parameters, which are elaborated in section 6 of the paper, and were calculated using 5 fold cross validation.

Metrics (Optimized by SMOTE)	decision tree	logistic regression
Accuracy	0.828	0.804
F1 Score	0.499	0.487
ROC AUC	0.788	0.802
Class 0 Accuracy	0.841	0.805
Class 1 Accuracy	0.735	0.799
Optimized Parameters	'dt_classifier__max_depth': 10, 'dt_classifier__min_samples_leaf': 20, 'smote__k_neighbors': 7}	'log_reg_classifier__C': 1, 'smote__k_neighbors': 3
Top 3 Important Features	duration, campaign, housing	duration, housing, marital_married

Between the two models that were tuned, the best model ended up being the logistic regression one, which had a logistic regression of 0.799. However, both models greatly increased from the original benchmark 1, with the class 1 accuracy of the decision tree model increasing by more than 0.30 and the logistic regression model increasing by more than 0.60.

Feature testing rarely changed for the models from the previous set of SMOT models, other than the second feature in the decision tree classifier. Based on this, we can determine that for both models, the most important feature in predicting customer subscription is the duration of the marketing call made to the customer.

Pre and Post Processing Benchmarks

Overall, four main types of processing and evaluations were done on the dataset, which was then trained with both the decision tree Classification model and the logistic regression model. Two of these processes can be considered pre-processed models and are suitable for benchmark evaluations of the data. The other two, that introduce sampling and hyper parameter tuning. Below is a description of the processes conducted

- Benchmark 1 (Numerical): Any data variables that had non-numerical entries and part of the X training set were removed. Values in the y column were converted to 0 and 1, where 1 was a person who subscribed to the Portuguese banking service after the marketing calls.
- Benchmark 2 (Categorical Binning and Removal): In addition to the conversion of y to numerical columns, job occupations were simplified into one of 3 categories (unemployed, employed, and retired). All categorical entries that were “yes” and “no” were converted to 1/0. Auxiliary columns that were not directly related to churn was removed, and all rows that were missing entries were removed. All categorical variables that could not be simplified into binary values were converted using the One Hot Encoding method which converts categorical variables into numerical format via dummy variables.
- SMOTE Sampling: Starting with the data from Benchmark 2, SMOTE analysis was performed in order to resample the data so that the proportions of y were similar to each other. This allowed our models to focus more on customers who subscribed to the banking model, as well as provide more information about said customers and their preferences.
- SMOTE Hyperparameter Tuning; Additional tuning was conducted by SMOTE in order to determine what would maximize the most accurate prediction for customers that subscribed to the model.

Comparing All Models

The model that ended up doing the best based on our goal of predicting subscriptions was the logistic regression model that used SMOTE and had hyperparameters tuned. This makes sense as SMOTE sampling balanced out the data we used, and hyperparameter turning made sure that parameters that the models used were optimized based on class 1 accuracy.

An interesting development we noticed was that both of the benchmark models were similar to each other, even though the first model contained less features than the second model. Additionally, though our benchmark models were performing better with the decision tree classifier, the model that did the best for the data was the logistic regression model, which did worse on the class 0 accuracy and f1 score, but did better on the ROC AUC and class 1 accuracy scoring.

Overall the duration of the call was consistently the factor that determined subscription rate, which was common among all models and at all forms of processing. Additional factors that were considered were often related to customer statistics, such as account balance, marital status, and current housing of the person contacted, as well as if they were contacted previously.

Helpful Preprocessing Steps

Accuracy is extremely important when wanting to obtain optimization within a particular marketing campaign. Though we took quite a few preprocessing steps, certain variables did not have as big an impact compared to other variables which created a better accuracy. Below are listed our most valuable preprocessing steps that would be crucial in skipping:

Removing any data leakage:

The removal of 'duration' was by far the most valuable step to take in order to turn both our models into something realistic and usable. Although it originally can be seen to produce higher overall scores, this is not given until after a call ends which defeats the purpose of our desire to know who we should reach out to call. Seeing that our F1 Score dropped significantly after this removal, we were able to recognize the leakage occurring due to this variable. The overall outcome for this removal helped increase model performance and ensure that it is much more honest and reliable.

Proper encoding and cleaning of variables:

The process of properly encoding variables in which we eliminated N/A variables, grouped job titles as a one, and performed 'One Hot Encoding' greatly helped in the organization of outcomes ensuring they would be results which did not generate errors, confusion, or any unidentifiable variables. In performing this cleaning process, we ensured that all of our machine learning models would be able to interpret categories directly, get rid of the extra noise with N/A variables, and prevent models from incorrectly treating job categories as numerical magnitudes. Had we not done this process, our models, most specifically the logistic regression model, would have failed to produce any valid calculations. Our data would have caused there to be poor performance or fail to converge.

Sampling

Since the data had a disproportionate amount of 0s, or people that did not subscribe to the banking institution, to 1s, or people that did, any models that were made on the data would optimize 0's or what customers the marketing strategy would appeal to. Because we care about factors of the customers that would lead to increased subscription, sampling, or picking the data

so the proportions of 0s and 1s were balanced was important in order for the model to find trends related to people who did subscribe. SMOTE did this by sampling the data so that the number of 0s and 1s would be similar, and helped us get trends based on the data that we wanted to learn about.

Hyperparameter Tuning

In order to ensure that our decision tree did not grow too deep, obtain noise, or fail to split, we tuned several key variables that will allow us to prohibit there to be any sort of overfitting. As a result of tuning, we were able to improve generalization and produce the best test F1.

Key factors modified ones such as:

- Max_depth: which controlled how deep the tree would grow
 - Tuned in order to find a balance depth that captures signal without memorizing any outside irrelevant noise.
- Min_samples_split: specific samples required in order to perform a split
 - Too small of a value will result in an aggressive overfitting. Larger values will force the module to only make splits when there is enough evidence given. This will smooth decision boundaries while improving stability
- Min_samples_leaf: stabilizes leaves while reducing noise
- Criterion: Selects best impurity measure
- Logistic Reg Classifier: Controlled how fit the model was to the data
- KNN for Smote: How the data that was generated was created (based on how many closest points)
- Class weight: Helps compensate for any imbalance in the class

6. Key Takeaways

In our entire analysis and experiments performed, we were able to learn several important concepts when it came to data mining and usage of machine learning methods.

Data Quality

Having clean data for our models is the most important factor determining our results. The cleaning of missing values, combining employment types into one category, as well as eliminating any leakage such as 'duration' created a bigger impact with more accurate results in realistic scenarios such as the one we are given.

Imbalanced Datasets Require Careful Handling

While our goal is to increase the 'yes' responses from customers, we were not fortunate enough to get a balanced dataset. As always, we are up for the challenge which is why we originally chose this dataset which contained only 11% of 'yes' responses. Without any balanced strategies, models tend to predict 'no' for nearly every observation.

Hyperparameter Tuning and Processing

Hyperparameter tuning is an important technique to calculate the most effective setting for our models, and often uses statistical calculations that can be missed with only human data analysis.

In order to produce the most accurate models, data not only needs to be collected well, but also have the right parameters to process it. Additional work such as pruning and forms of regression are also helpful to know about in order to improve model accuracy.

Advice for Business Managers

In order to reach full optimization in this telephone marketing campaign, immediate action should be taken with the predictors addressed. In order to ensure there is a great return on the money spent for the campaign, efforts to contact disengaged customers must be eliminated. Dedication any time to customers with low likelihood of subscribing will just waste more of the budget and time. Instead, prioritize customer segments who share similar characteristics such as: retired, higher level education standing, no personal loans, and positive previous campaign results. Additionally, when contacting customers, be mindful of the duration of the call, since it might convince or dissuade them from returning to the product.

Additionally, focus on the jobs, marital status, balance in the account, and housing status of the customer when tailoring advertising methods. Retired customers will want to hear about the security regarding this investment, higher education customers will want proof and understanding of the long term growth with this investment, and lastly, customers who have been repeatedly contacted should be avoided and no longer contacted as this consistently showed a result of customers answering 'no' at the end. Instead, set a limit to 1-2 attempts max on contacting customers.

Other Recommendations

Though we were able to create a high accuracy analysis, there are a few recommendations that should be applied in order to improve future marketing campaigns.

Implementation of Predictive Targeting System

To simplify the targeting process for the next campaign, integrate a decision tree into the marketing workflow in order to better generate a split of high and low probability customers. This will help increase subscriptions as well as greatly reduce unnecessary calls.

Collect Informative Data:

Even though this may be categorical data, tracking reason of refusals, customer sentiment, and preferred contact times can create a better more fluent campaign that will ensure improvement of predictive accuracy. Having the understanding of why customers responded the way they did can help build more accurate models going forward and reduce wastefulness in the campaign.

Limitations

In doing our analysis we are cognizant of limitations in the data and our analytical process. The data we used is from Portugal in 2008. This means that banking trends that were common at the time may not be relevant in 2025, or even in the United States based on business and market

standards that the data was collected under. For example, many forms of marketing come in the form of ads on the internet or social media, and extensive spam programs catch most phone and email marketing campaigns. Additionally, our data was limited since we were using mostly categorical data in order to create the models, which we often needed to simplify and convert to numerical data in order to analyze. In terms of our analysis, we used the SMOTE data sampling procedure to synthetically generate new data points, which means that not all data used at the end was taken from customers that were called, and some data points may not have been included in the data that was used in making the model, which can lead to inaccurate results.

Technical Appendix

Data Set: <https://archive.ics.uci.edu/dataset/222/bank+marketing>

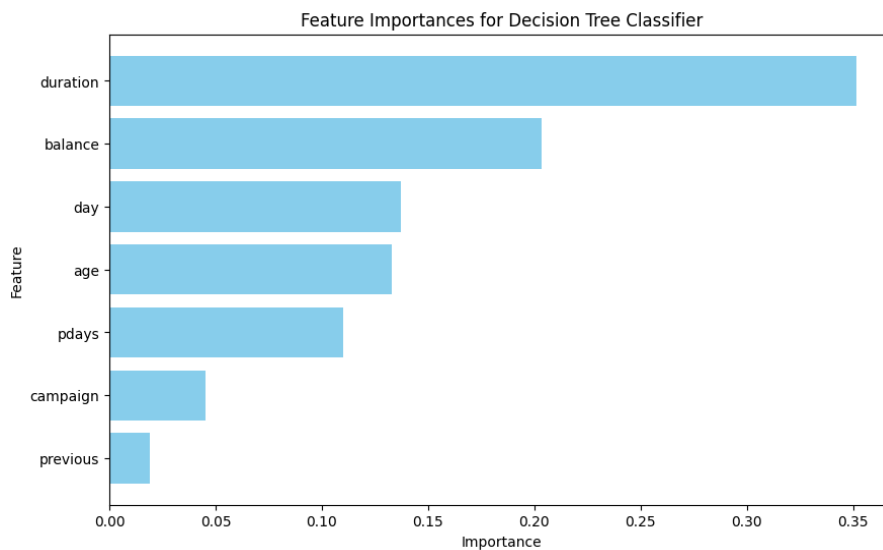
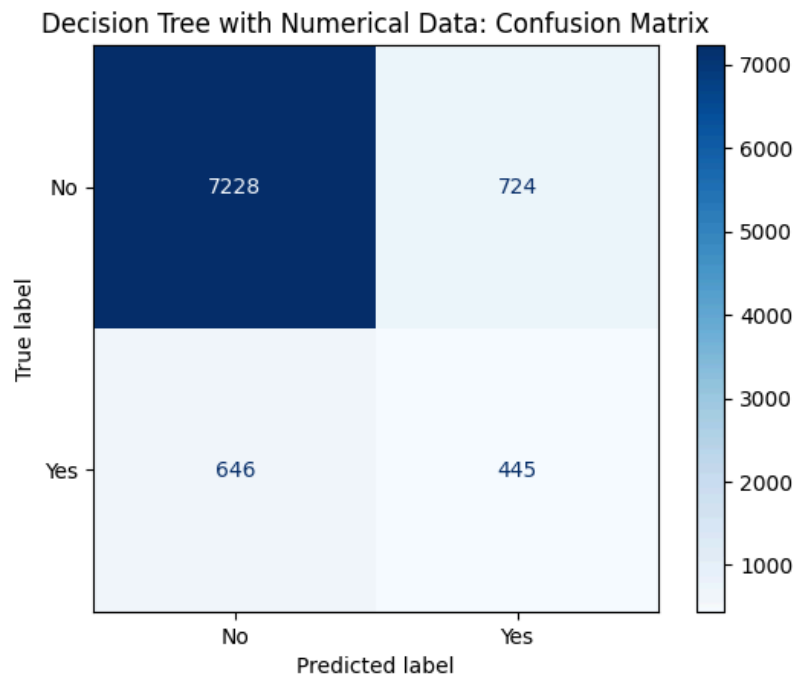
Google Colab:

<https://colab.research.google.com/drive/1hDadJ2S8Y25r5SaDhaTGS7qVz6vLzW3f?usp=sharing>

Visual Models

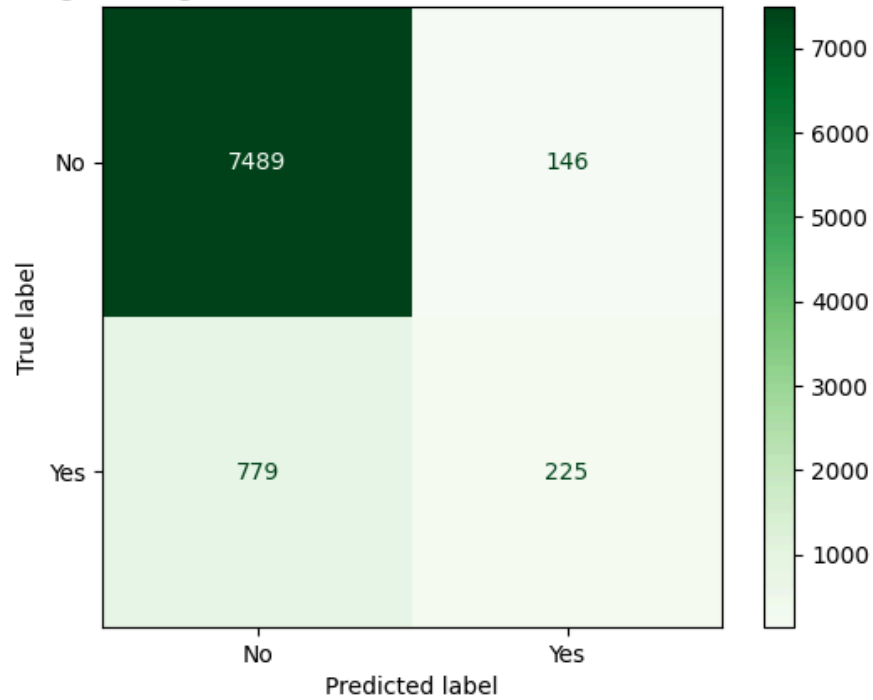
Benchmark 1: Numerical Data

Decision Tree Classification

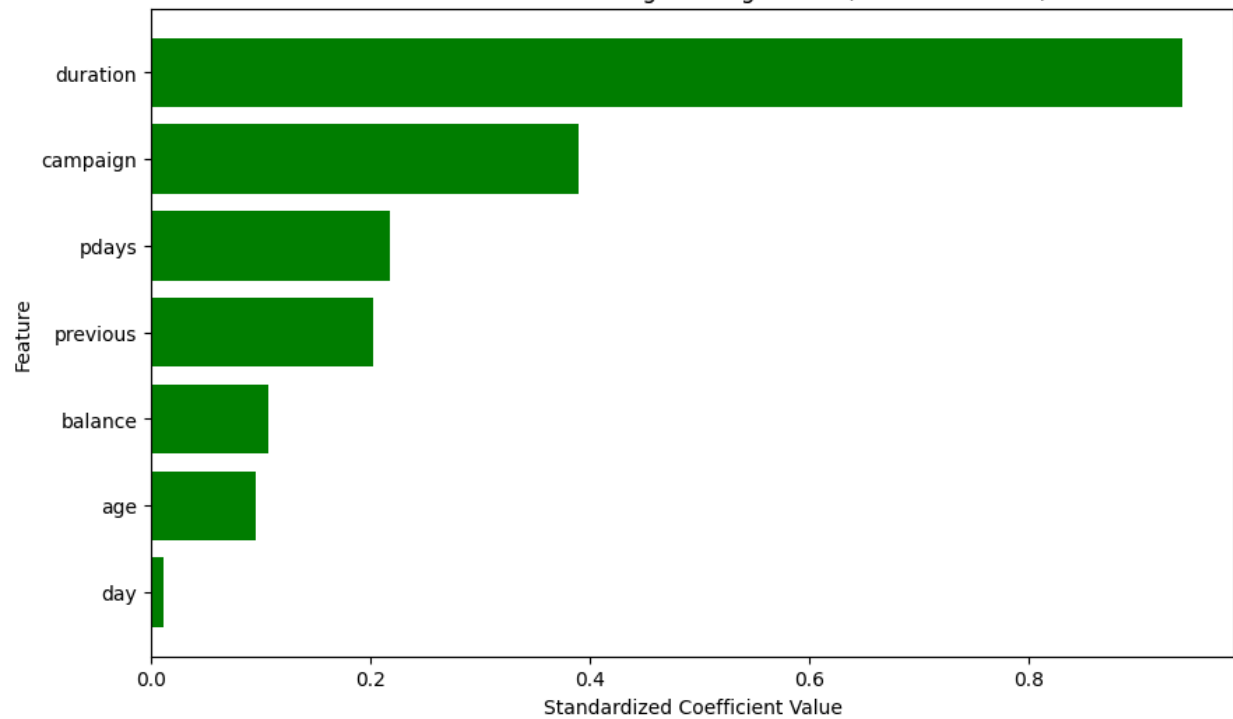


Logistic Regression

Logistic Regression: Confusion Matrix (Scaled Features)



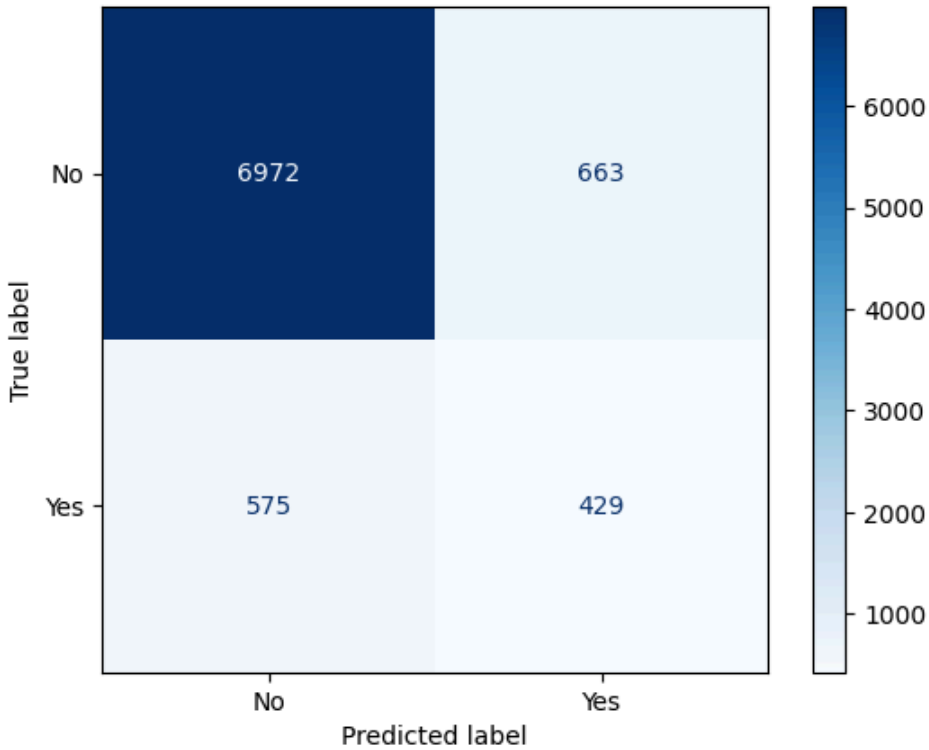
Feature Coefficients for Logistic Regression (Scaled Features)



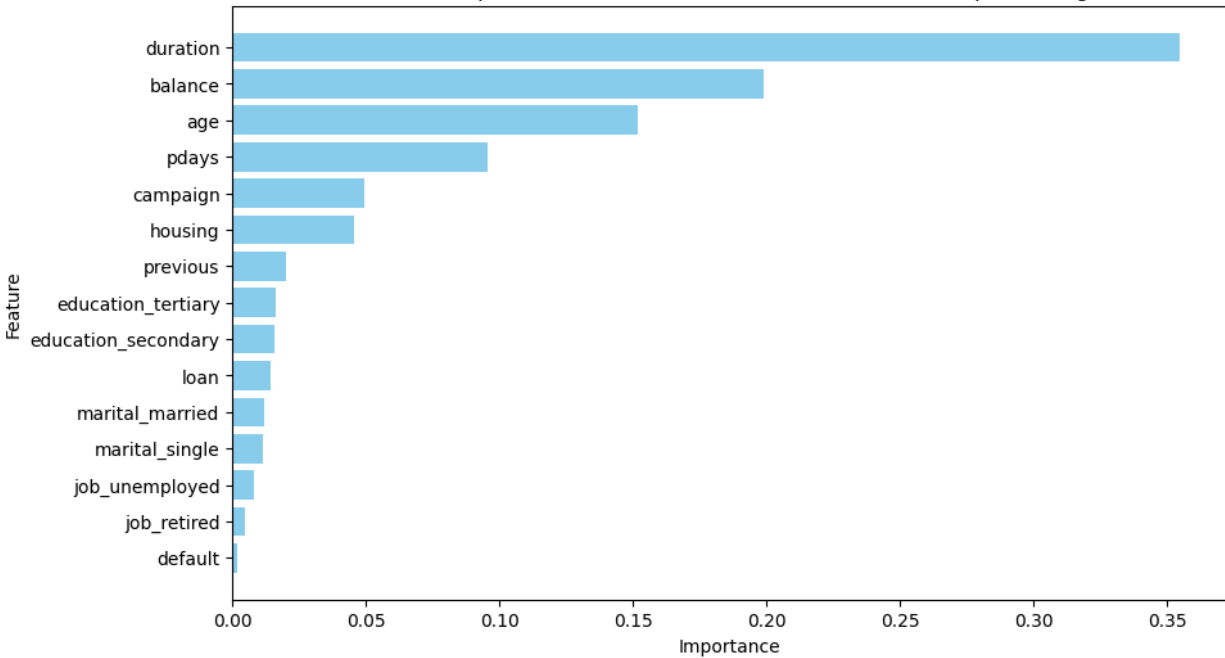
Benchmark 2: Data Cleaning

Decision Tree Classification

Decision Tree with Preprocessing: Confusion Matrix

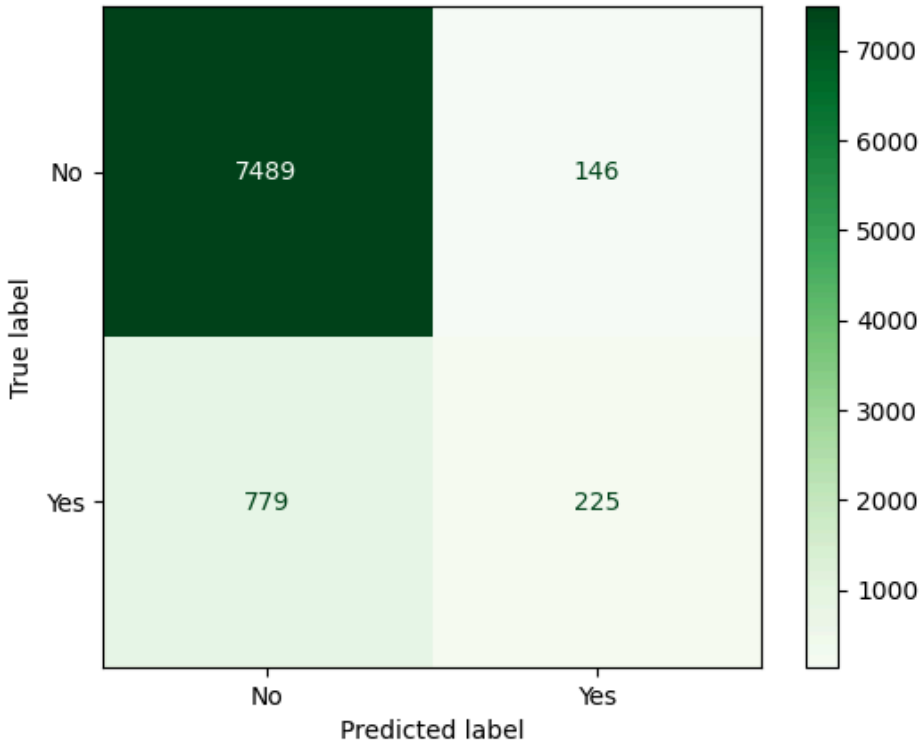


Feature Importances for Decision Tree Classifier (with Preprocessing)

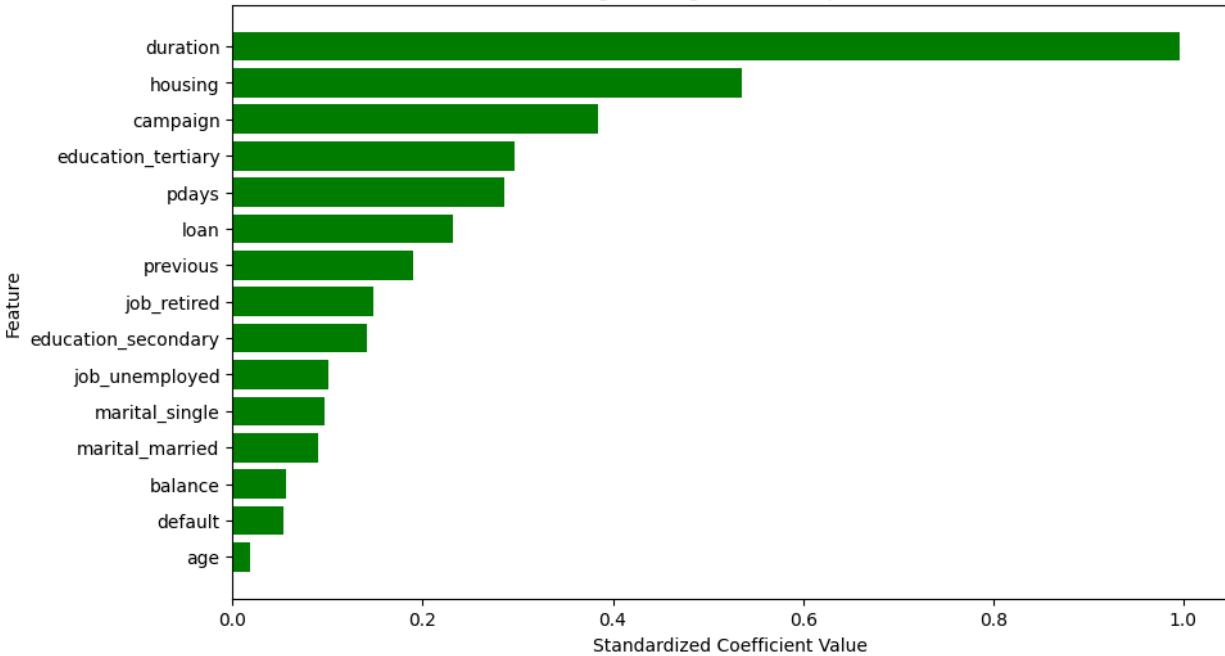


Logistic Regression

Logistic Regression with Preprocessing: Confusion Matrix

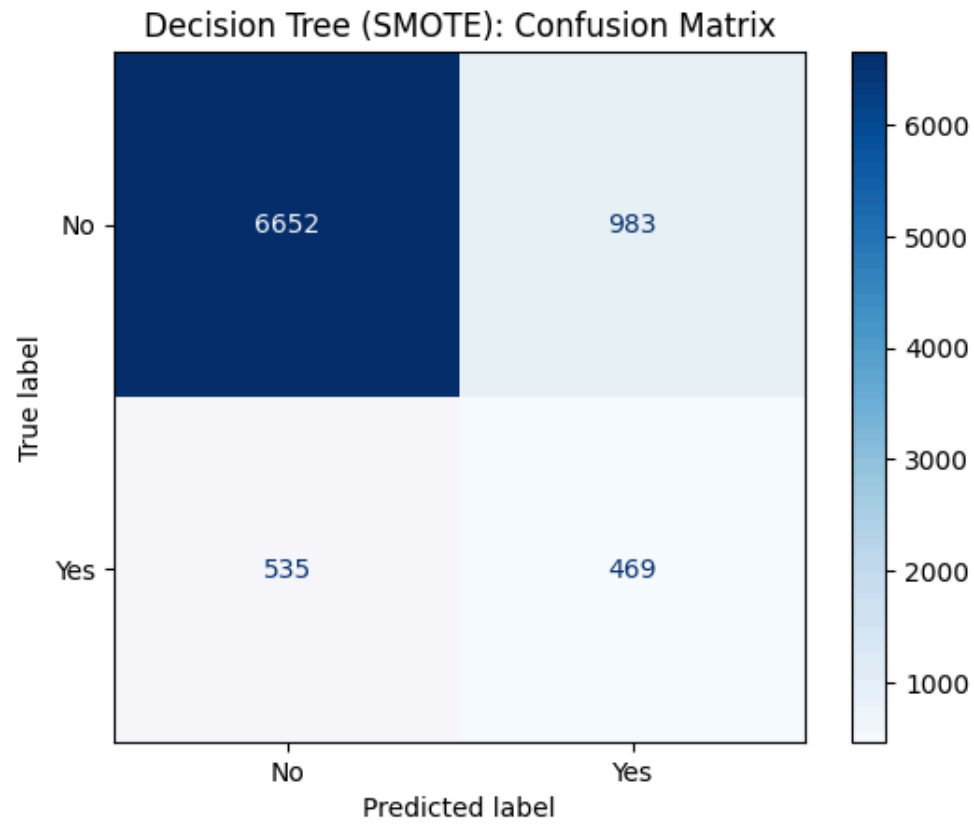


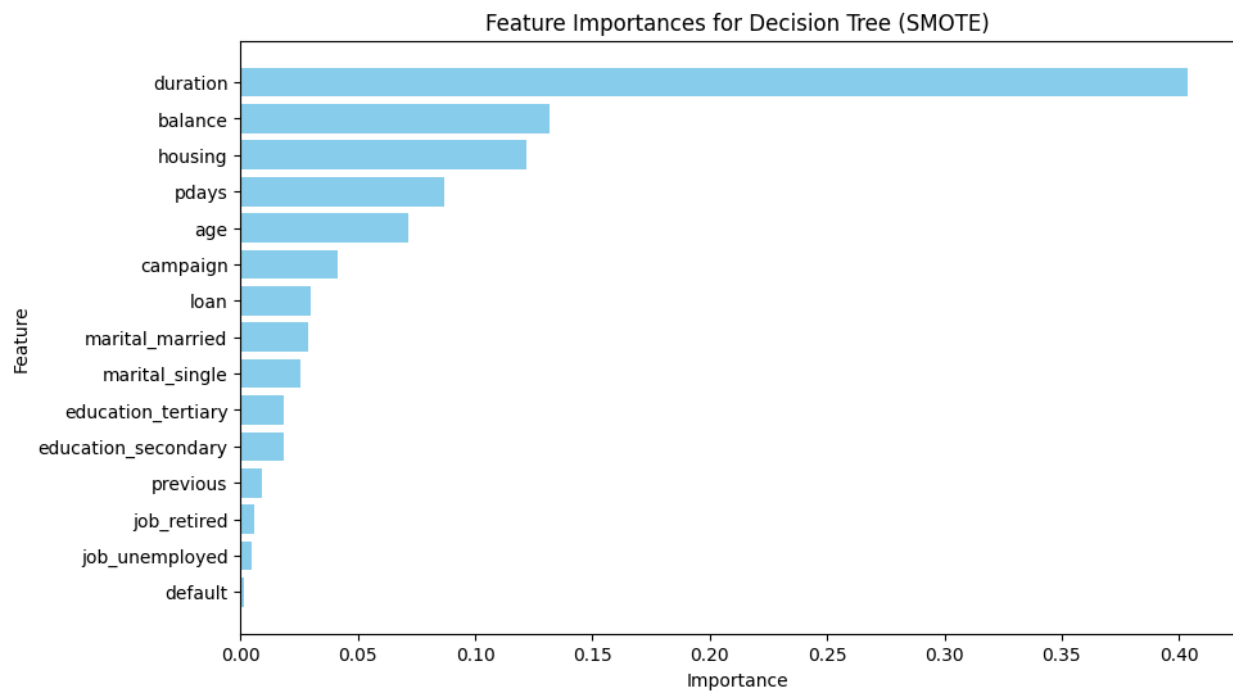
Feature Coefficients for Logistic Regression (Preprocessed & Scaled Features)



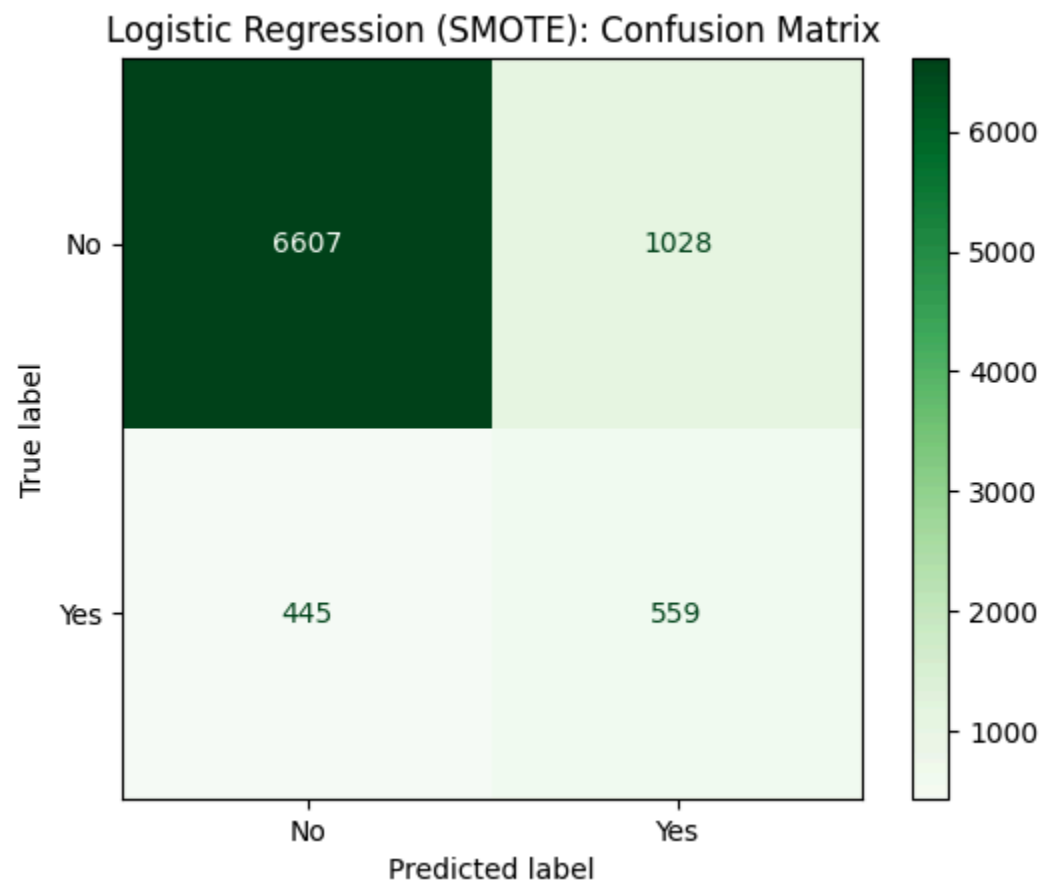
SMOTE Sampled Data

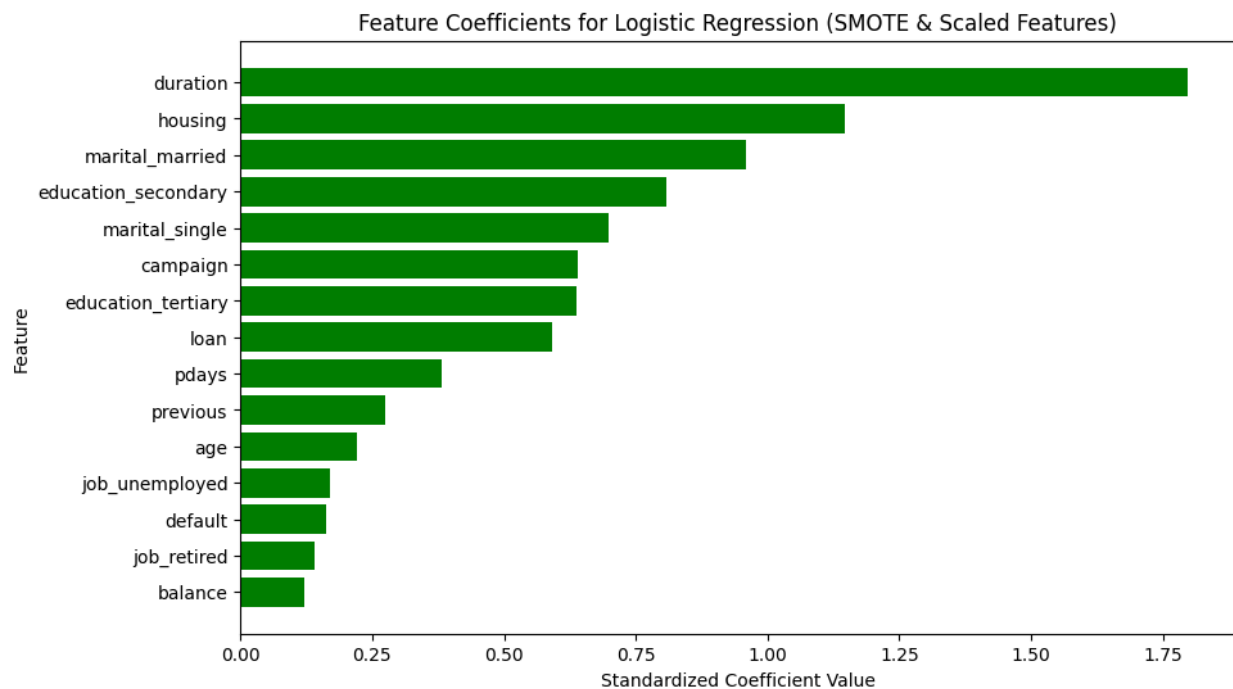
Decision Tree Classification





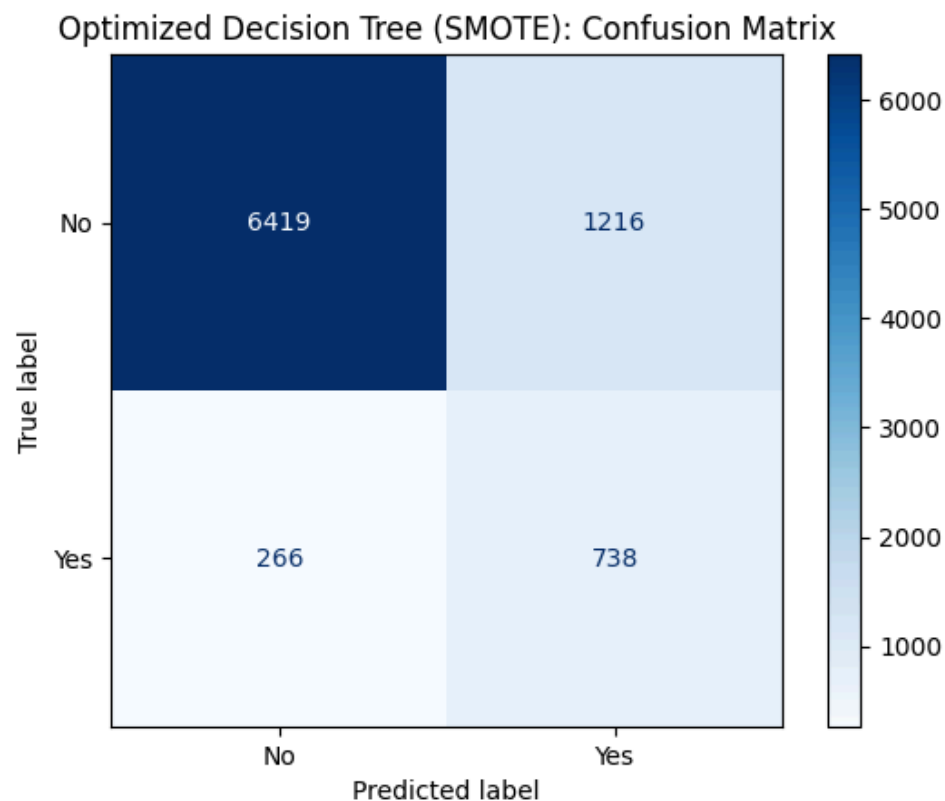
Logistic Regression

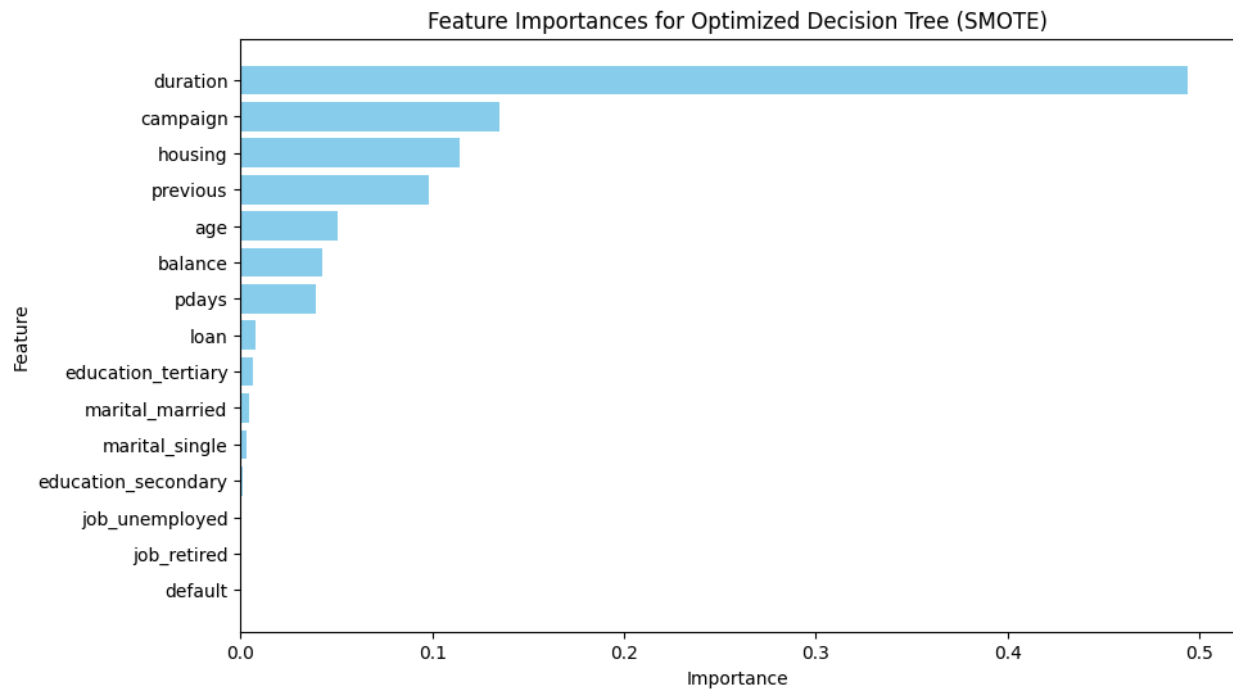




SMOTE Sampled + Hyperparameter Optimization

Decision Tree Classification





Logistic Regression

Optimized Logistic Regression (SMOTE): Confusion Matrix

